

Performance Testing

Date	15 July 2026
Team ID	SWTID-2026-5339
Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the responsiveness, stability, and scalability of the EduGenie platform under different user loads. The objective was to ensure that the application performs efficiently while multiple students access learning resources, quizzes, AI chatbot, and study materials simultaneously.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Student Login, AI Chatbot, Quiz Module, Study Material Access
Test Environment	Localhost (Flask Application)
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Student Login	20	60	Users should log in successfully without delay
2	AI Chatbot Response	15	60	Chatbot should respond within 2 seconds
3	Quiz Submission	25	90	Quiz should be submitted successfully
4	Study Material Access	30	60	Learning resources should load without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 sec	1.3 sec	Pass	Fast response
2	Response Time (Max)	< 5 sec	3.4 sec	Pass	Within acceptable limit
3	Throughput	100 Req/sec	112 Req/sec	Pass	Good performance
4	Error Rate	< 1%	0.2%	Pass	Very low error rate
5	CPU Utilization	< 80%	58%	Pass	Stable resource usage
6	Memory Utilization	< 80%	64%	Pass	Memory usage within limit

Step 4: Observations & Analysis

- EduGenie handled multiple concurrent users efficiently.
- Login, quiz, and study material modules responded quickly.
- The AI chatbot generated responses within the expected response time.
- No significant server crashes or performance degradation were observed during testing.
- Overall, the application demonstrated stable and reliable performance under the tested load.

Step 5: Screenshots / Evidence

Attach screenshots of the performance testing results, such as:

- Apache JMeter Test Plan
- Summary Report
- Aggregate Report
- Response Time Graph
- Throughput Graph

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]